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Primer - Analytics SDK to Enterprise SDK Migration

The WHY?

FactSet will **deprecate Analytics SDKs (Python, Java, and .Net)** effective **31-Aug-2026.** This change supports FactSet's ongoing efforts to modernize and enhance its technological ecosystem. Transition to the **FactSet Enterprise SDKs** to maintain continued support, performance improvements, and security updates.

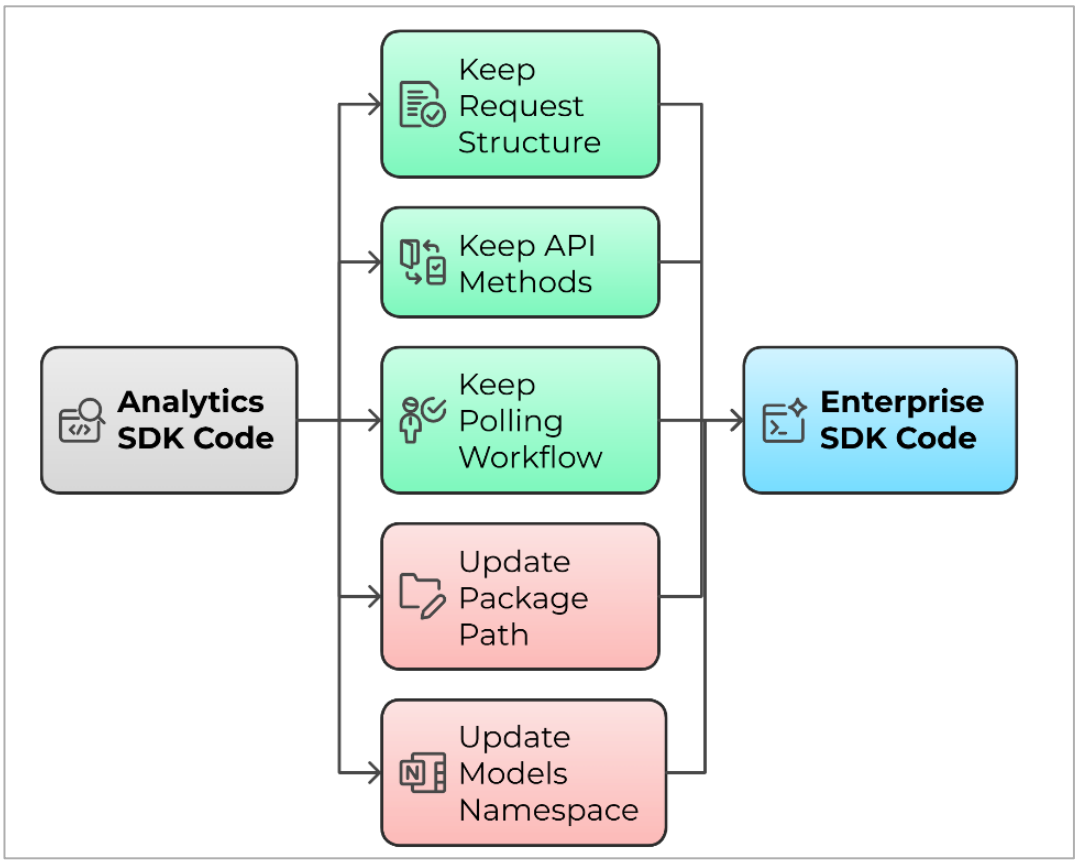
The HOW?

Migration guides are available for Python, Java, and .NET on the Developer Portal to support the transition to Enterprise SDKs:

- Python Migration Guide
- Java Migration Guide
- .Net Migration Guide

Additionally, you can refer to the instructions below and the Migration Demo Video for quick, step-by-step guidance on the migration changes.

The SDK migration process across the same API version is straightforward, **only the package path and models namespace change:**



Analytics SDK and Enterprise SDK Differences

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NOTE: If you are using version 2 of the Engine APIs, refer to the migration guides to upgrade to version 3 and migrate to the Enterprise SDKs. For more details regarding v2 to v3 migration, please see:

- v2 to v3 Migration One Pager
- v2 to v3 Migration Detailed Doc

ACTION ITEMS - Checklist and Details

1. Install the Enterprise SDK
2. Update imports (main migration change)
3. Run the calculation, Poll status and retrieve result
4. Convert STACH results to tables/dataframes

1. Install the Enterprise SDK

Code Updates	Python	Java	.NET
Source	Python source	Java source	.NET source
Utilities	Python utilities	Java utilities	.NET utilities

FactSet Enterprise SDK Overview

2. Update Imports (Analytics SDK vs Enterprise SDK)

OLD (Analytics SDK)

NEW (Enterprise SDK)

Python

```python
import time
import os
import uuid
import pandas as pd
import urllib3

from fds.analyticsapi.engines import ApiException
from fds.analyticsapi.engines.api.pa_calculations_api import PACalculationsApi
from fds.analyticsapi.engines.api.components_api import ComponentsApi
from fds.analyticsapi.engines.api.pricing_sources_api import PricingSourcesApi
from fds.analyticsapi.engines.api_client import ApiClient
from fds.analyticsapi.engines.configuration import Configuration
from fds.analyticsapi.engines.model.component_summary import ComponentSummary
from fds.analyticsapi.engines.model.pa_calculation_parameters_root import PACalculationParametersRoot
from fds.analyticsapi.engines.model.pa_calculation_parameters import PACalculationParameters
from fds.analyticsapi.engines.model.pa_date_parameters import PADateParameters
from fds.analyticsapi.engines.model.pa_identifier import PAIdentifier
from fds.analyticsapi.engines.model.pa_calculation_data_sources import PACalculationDataSources
from fds.analyticsapi.engines.model.pa_calculation_pricing_source import PACalculationPricingSource
from fds.protobuf.stach.extensions.StachVersion import StachVersion
from fds.protobuf.stach.extensions.StachExtensionFactory import StachExtensionFactory
from fds.protobuf.stach.extensions.v2.StachUtilities import StachUtilities
```

```python
import time
import os
import uuid
import pandas as pd

from fds.protobuf.stach.extensions.StachVersion import StachVersion
from fds.protobuf.stach.extensions.StachExtensionFactory import StachExtensionFactory

from fds.sdk.PAEngine.api.components_api import ComponentsApi
from fds.sdk.PAEngine.api.pricing_sources_api import PricingSourcesApi
from fds.sdk.PAEngine.apis import PACalculationsApi
from fds.sdk.PAEngine import ApiClient, Configuration, ApiException
from fds.sdk.PAEngine.models import *
from fds.sdk.utils.authentication import ConfidentialClient
```

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.NET

```dotnet
using System;
using System.Collections.Generic;
using System.Linq;
using System.Net;
using System.Threading;
using System.Diagnostics;

using FactSet.AnalyticsAPI.Engines.Api;
using FactSet.AnalyticsAPI.Engines.Client;
using FactSet.AnalyticsAPI.Engines.Model;

using FactSet.Protobuf.Stach.Extensions;
```

```dotnet
using System;
using System.Collections.Generic;
using System.Linq;
using System.Net;
using System.Threading;
using System.Threading.Tasks;
using FactSet.Protobuf.Stach.Extensions;
using FactSet.SDK.PAEngine.Api;
using FactSet.SDK.PAEngine.Client;
using FactSet.SDK.PAEngine.Model;
using FactSet.SDK.Utils.Authentication;
```

Java

```java
package examples;

import java.io.File;
import java.io.FileOutputStream;
import java.util.List;
import java.util.Map;
import java.util.UUID;

import javax.ws.rs.client.ClientBuilder;

import com.factset.protobuf.stach.extensions.models.Row;
import com.factset.protobuf.stach.extensions.v2.StachUtilities;
import com.google.protobuf.InvalidProtocolBufferException;
import com.google.protobuf.Value;
import org.glassfish.jersey.apache.connector.ApacheConnectorProvider;
import org.glassfish.jersey.client.ClientProperties;

import factset.analyticsapi.engines.*;
import factset.analyticsapi.engines.api.*;
import factset.analyticsapi.engines.models.*;

import factset.analyticsapi.engines.models.CalculationStatus.StatusEnum;

import com.factset.protobuf.stach.extensions.ColumnStachExtensionBuilder;
import com.factset.protobuf.stach.extensions.RowStachExtensionBuilder;
import com.factset.protobuf.stach.extensions.StachExtensionFactory;
import com.factset.protobuf.stach.extensions.StachExtensions;
import com.factset.protobuf.stach.extensions.models.StachVersion;
import com.factset.protobuf.stach.extensions.models.TableData;
import com.fasterxml.jackson.core.JsonProcessingException;
import com.fasterxml.jackson.databind.ObjectMapper;

import org.apache.poi.xssf.usermodel.XSSFCell;
import org.apache.poi.xssf.usermodel.XSSFRow;
import org.apache.poi.xssf.usermodel.XSSFSheet;
import org.apache.poi.xssf.usermodel.XSSFWorkbook;
```

```java
package examples;

import com.factset.sdk.PAEngine.*;
import com.factset.sdk.PAEngine.api.*;
import com.factset.sdk.PAEngine.models.*;
import com.factset.sdk.utils.authentication.ConfidentialClient;

import com.factset.protobuf.stach.extensions.RowStachExtensionBuilder;
import com.factset.protobuf.stach.extensions.StachExtensionFactory;
import com.factset.protobuf.stach.extensions.StachExtensions;
import com.factset.protobuf.stach.extensions.models.StachVersion;
import com.factset.protobuf.stach.extensions.models.TableData;
import com.factset.protobuf.stach.extensions.models.Row;
import com.factset.protobuf.stach.extensions.v2.StachUtilities;

import com.factset.sdk.utils.exceptions.AuthServerMetadataContentException;
import com.factset.sdk.utils.exceptions.AuthServerMetadataException;
import com.factset.sdk.utils.exceptions.ConfigurationException;
import com.fasterxml.jackson.core.JsonProcessingException;
import com.fasterxml.jackson.databind.ObjectMapper;
import com.google.protobuf.InvalidProtocolBufferException;
import com.google.protobuf.Value;
import org.apache.poi.xssf.usermodel.XSSFCell;
import org.apache.poi.xssf.usermodel.XSSFRow;
import org.apache.poi.xssf.usermodel.XSSFSheet;
import org.apache.poi.xssf.usermodel.XSSFWorkbook;
import org.glassfish.jersey.apache.connector.ApacheConnectorProvider;
import org.glassfish.jersey.client.ClientProperties;

import javax.ws.rs.client.ClientBuilder;
import java.io.File;
import java.io.FileOutputStream;
import java.security.KeyManagementException;
import java.security.NoSuchAlgorithmException;
import java.util.List;
import java.util.Map;
import java.util.UUID;
```

3. Run the calculation → Poll Status → Retrieve Result

Keep the same v3 request body, top-level data wrapper stays as is. It is a v3 API construct and not an SDK difference.

Call the same methods as before to run/update the calculation, to poll the status of the calculation and retrieve the result once the calculation is completed. Refer to the table below for methods and details for each API:

APIs	Python	Java	.NET
PA Engine API	PA_Python_Methods	PA_Java_Methods	PA_.NET_Methods
SPAR Engine API	SPAR_Python_Methods	SPAR_Java_Methods	SPAR_.NET_Methods

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Vault API	Vault_Python_Methods	Vault_Java_Methods	Vault_.NET_Methods
Quant Engine API	Quant_Python_Methods	Quant_Java_Methods	Quant_.NET_Methods
Fixed Income Calculation API	FI_Python_Methods	FI_Java_Methods	FI_.NET_Methods

4. Converting STACH results to tables/dataframes

To receive calculation results as tables or dataframes, you must install the stach-extensions package before running calculations. In the new Enterprise SDK, stach-extensions is not included by default and must be installed separately. If it is not installed, results will be returned in STACH format instead of tables/dataframes.

GitHub Repo: https://github.com/factset/stach-extensions

	Python	Java	.NET
Installation Code for STACH Extensions	pip install fds.protobuf.stach.extensions	Add the below dependency to the project: <dependency> <groupId>com.factset.protobuf</groupId> <artifactId>stachextensions</artifactId> <version>ARTIFACT_VERSION</version> </dependency>	Install with Package Manager Console: Install-Package FactSet.Protobuf.Stach.Extensions Install with NuGet: nuget install FactSet.Protobuf.Stach.Extensions Install with .NET Core: dotnet add package FactSet.Protobuf.Stach.Extensions

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